

EDUCATION

Charlottesville, VA

University of Virginia

Aug 2016 - May 2020

- **Major:** Computer Science, B.A.
- **Minors:** Economics, Astronomy

EMPLOYMENT

Twilio - *Senior Software Engineer*

July 2022 - Present

- As a member of TaskRouter, scaled from 50M to 120M+ monthly tasks while maintaining 99.999% uptime, transforming it into one of Twilio's most reliable and high-traffic systems powering the Flex platform, including critical emergency and healthcare services.
- Designed and implemented a robust rate-limiting solution adopted across Twilio's core library, now widely used by multiple teams and regularly consulted on implementation two years post-launch.
- Part of a leadership-initiated task force addressing a severe outage in the Conversations platform; identified and resolved Sev0 bugs, implemented key enhancements, and helped restore customer confidence. Now driving backend modernization to scale from thousands to millions of MAUs.
- Regularly lead high-pressure on-call rotations overseeing millions of real-time customer interactions, including emergency and healthcare services. Diagnosed issues under pressure and implemented post-incident improvements to drive long-term system resilience.
- Promoted to Senior Engineer at age 25, one of the youngest among 1,500+ engineers across Twilio.

Goldman Sachs - *Software Engineer*

May 2021 – July 2022

- As a member of the Inventory Management team, built trading and inventory-management logic for Goldman Sachs' repo desk, systemizing validation and auto-execution of thousands of daily security movements (billions in firm inventory) directly impacting desk liquidity and P&L.
- Promoted within 8 months and entrusted to design and implement Flow Type Auto Execution Thresholds — a new control mechanism improving the safety and efficiency of repo trade validation.
- Worked with leadership in Goldman's Treasury to implement specific requirements in Delphi's human review pipeline for select repo allocations.
- Integrated Delphi with a new downstream API to connect to a new source of U.S. Treasuries held by the firm, increasing available liquidity and boosting desk P&L.
- Served in high-pressure 24-hour on-call rotations supporting sub-24-hour repo allocations, resolving time-sensitive issues that directly affected Goldman's revenue.

Fannie Mae - *Software Engineer*

June 2020 - May 2021

- As a member of the Financial Reporting team, managed and extracted highly sensitive data from company databases to build complex financial reports used throughout all levels of the business.

RESEARCH & PROJECTS

Macro Economic Research and Projects: <https://github.com/adssoccer1/Macro>

- **Series 3** – National Commodities Futures Examination, FINRA — *Passed August 2024*
- Produced data-driven research combining statistical analysis, historical backtests, and real-time market data on U.S. monetary/fiscal policy, yield-curve trades, FX strategy, and global trade flows.
- Developed Python-based models quantifying equity, rates, FX correlations, enabling systematic testing of macro theses and portfolio strategies.

Skills

- **Finance & Markets:** Macro trading, STIR (SOFR), U.S. yield curve, FX strategy, systematic strategies
- **Software Engineering:** Java, Python, SQL, AWS (S3, SQS, DynamoDB), distributed systems, real-time messaging, Kafka, C++, Git, Javascript, React